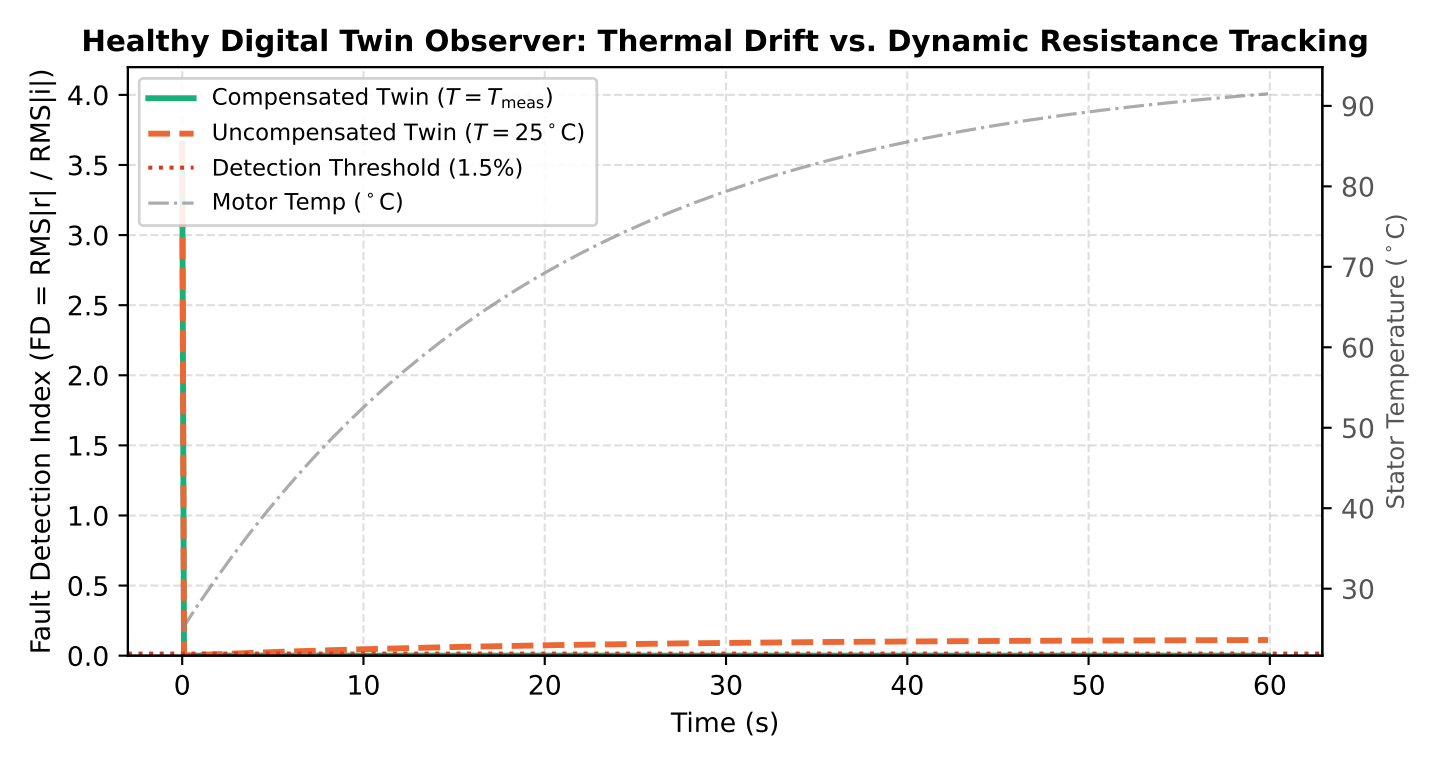
Healthy Digital Twin Observer: Thermal Drift vs. Dynamic Resistance Tracking
Compensated Twin ($T = T_{meas}$)
Uncompensated Twin ($T = 25\,^\circ$C)
Detection Threshold (1.5%)
Motor Temp ($^\circ$C)
Fault Detection Index (FD = RMS|r| / RMS|i|)
Stator Temperature ($^\circ$C)
Time (s)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
30
40
50
60
70
80
90
0
10
20
30
40
50
60